FIG. 8B -- Gossip Replication Protocol (Page 2 of 2) 8/16

== SECTION D: DEDUPLICATION 806 ==

```
Bounded Deduplication Set 806

seenMessages: string[] (ordered, oldest first)
seenMessageSet: Set<string> (O(1) lookup)
MAX_SEEN_MESSAGES = 10,000

markSeen(messageId):
   if set.has(id): return (already tracked)
   if length >= 10,000: evict oldest 5,000 (half), rebuild Set
   push(messageId)    +    set.add(messageId)

Before eviction: After eviction:
[msg_1 ... msg_5000 ... [msg_5001 ... msg_10000, NEW_MSG]
 msg_10000] ^new oldest newest^
 ^oldest newest^
```

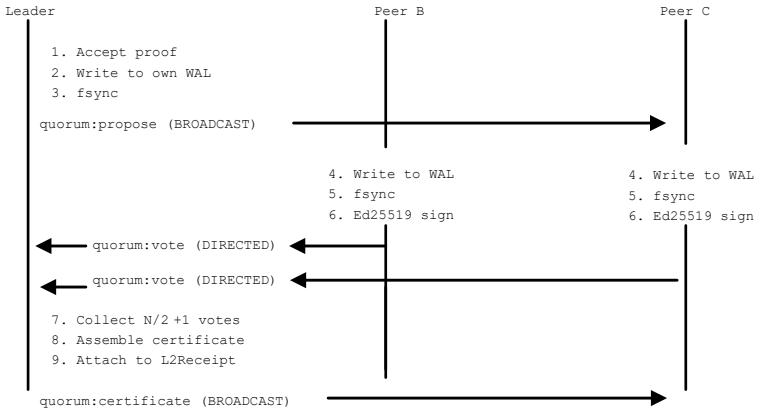
== SECTION E: BROADCAST vs DIRECTED MESSAGING 808 ==

BROADCAST (11 types) DIRECTED (1 type)

```
sender.broadcast(msg)
-> ALL connected peers
-> each relays (except sender)

Types: proof:accepted,
   batch:built/settled/failed,
   message:stored, state:snapshot,
   leader:heartbeat/promotion,
   quorum:propose/certificate,
   peer:announce
```

```
sender.sendTo(peerId, msg)
-> ONE specific peer, NO relay

Type: quorum:vote ONLY
Why: vote goes BACK to proposing
leader collecting N/2    +    1 votes.
Broadcasting would waste
bandwidth.

sendTo() returns false if peer
not found/disconnected.
```

== QUORUM DURABILITY FLOW (3-MESSAGE SEQUENCE) ==



Threshold: N/2 +1 (e.g., 2 of 3) Timeout: 5,000ms (proceed without cert)
Each vote = Ed25519 signature over canonical pipe-delimited message.

== TRUST MODEL ==

GOSSIP PROVIDES:

```
Replication (all nodes get data)
Durability (WAL    +     quorum acks)
Leader discovery (heartbeats)
Failover (promote)
Peer discovery (announce)
Auth (HMAC)    +    Non-repudiation
   (Ed25519)
```

GOSSIP DOES NOT PROVIDE:

```
Proof validity (ZK crypto only)
Ordering consensus (partition
   math is deterministic)
Byzantine fault tolerance
Proof of work / proof of stake
Validity-determining voting
```

Proof pathway: ZK proof on device -> partition leader -> o1js verify() ->
   if valid: accept+ gossip -> ALL replicas trust (no re-verification)

== NETWORK PARAMETERS ==

```
MAX_SEEN_MESSAGES: 10,000          | Eviction: half (oldest 50%)
DEFAULT_MAX_PEERS: 12              | Peer announce: 60s
Heartbeat interval: 5,000ms        | Heartbeat timeout: 15,000ms
Quorum vote timeout: 5,000ms       | Quorum threshold: floor(N/2)           1
messageId: SHA-256 trunc 32 hex    | HMAC: HMAC-SHA256 (optional)
Vote sig: Ed25519 (128 hex)        | Delivery: HTTP via L2Client
```